

Chapter 33: Nervous Coordination: Nervous System and Sense Organs

Nervous System Overview

- The nervous system originated in the property of **irritability**, the ability to respond to environmental stimuli.
- **Neurons** (nerve cells) are the functional units of the nervous system.
- A typical neuron consists of:
 - **Cell body** (soma): Contains the nucleus.
 - **Dendrites**: Branched processes that receive information.
 - **Axon**: A single long process that carries signals away from the cell body.
- Axons are often covered with an insulating **myelin sheath** to accelerate signal propagation.
- Neurons are classified as:
 - **Afferent** (sensory): Carry signals to the central nervous system (CNS).
 - **Efferent** (motor): Carry signals from the CNS to effectors (muscles or glands).
 - **Interneurons**: Connect neurons within the CNS (99% of neurons in humans).



Figure 1: Structure of a motor (efferent) neuron. From the nucleated cell body extend cytoplasmic processes: dendrites and a single axon. The axon is often covered with an insulating myelin sheath formed by Schwann cells.

- **Nerves** are bundles of nerve processes (axons) in the peripheral nervous system.
- **Ganglia** are bundles of nerve cell bodies outside the CNS.



Figure 2: Structure of a peripheral nerve showing nerve fibers surrounded by various layers of connective tissue. A nerve may contain thousands of both efferent and afferent fibers bundled in fascicles.

- **Neuroglial cells** (glial cells) surround and support neurons.
 - **Schwann cells:** Form myelin in the peripheral nervous system.
 - **Oligodendrocytes:** Form myelin in the CNS.
 - **Astrocytes:** Provide nutrients and structural support; form the blood-brain barrier.
 - **Microglia:** Phagocytic cells involved in immune defense.

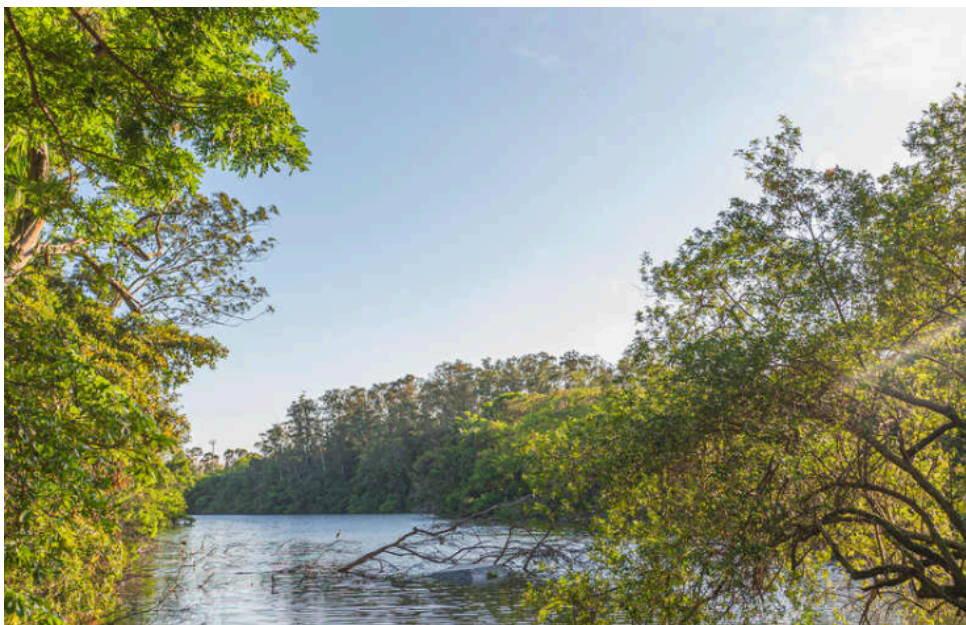


Figure 3: Development of the myelin sheath in a myelinated neuron of the peripheral nervous system. The Schwann cell grows around an axon, then rotates around it, enclosing the axon in a tight, multilayered sheath.

Nerve Action Potential

- A **nerve action potential** is an electrochemical message of neurons.

- It is an “all-or-none” phenomenon; signal intensity is conveyed by frequency.
- **Resting Membrane Potential:**
 - The membrane is selectively permeable to K^+ but nearly impermeable to Na^+ .
 - High concentration of Na^+ outside and K^+ inside.
 - The **sodium-potassium pump** maintains these gradients by pumping 3 Na^+ out and 2 K^+ in using ATP.
 - Resting potential is usually around -70 mV (inside negative).



Figure 4: Ionic composition inside and outside a resting nerve cell. The sodium-potassium pump drives sodium out and potassium in. The membrane is permeable to potassium but not sodium at rest, creating a resting potential where the inside is negative relative to the outside.

- **Action Potential:**
 - A rapid and brief **depolarization** of the membrane (reversal of polarity to $+35$ mV).
 - Caused by opening of voltage-gated Na^+ channels, allowing Na^+ influx.
 - **Repolarization** follows as Na^+ channels close and voltage-gated K^+ channels open, allowing K^+ efflux.
 - The action potential is self-propagating along the axon.



Figure 5: Conduction of an action potential. A: Originates in axon hillock. B & C: Electrical events and changes in membrane permeability. Sodium influx causes depolarization (inside becomes positive), followed by potassium efflux causing repolarization.

- **Conduction Velocity:**

- Depends on axon diameter (larger is faster) and myelination.
- In myelinated fibers, action potentials leap from node to node (**Nodes of Ranvier**).
- This is called **saltatory conduction** and is much faster than continuous conduction.

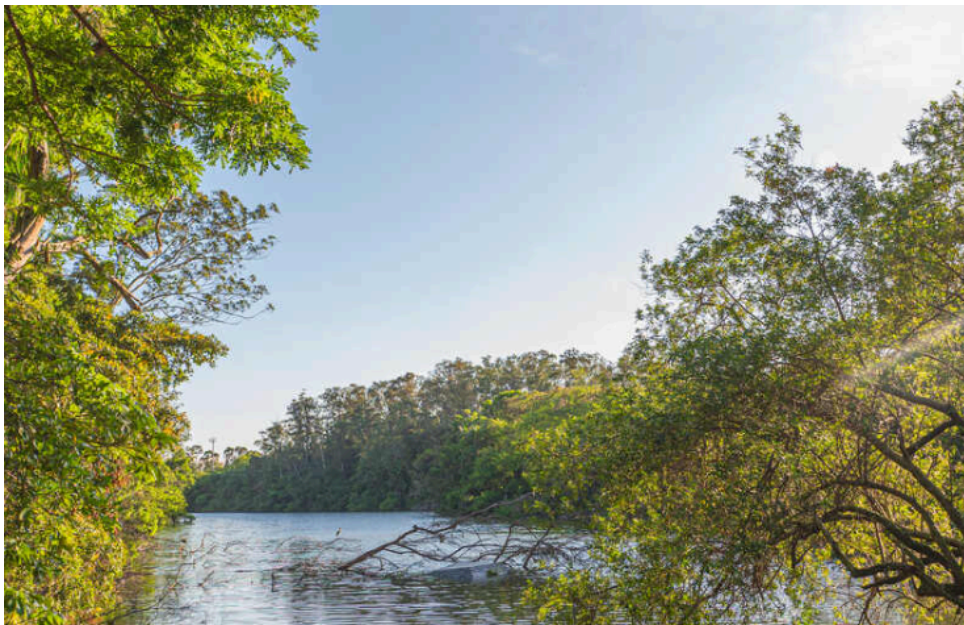


Figure 6: Action potential conduction in unmyelinated (A) and myelinated (B) fibers. In myelinated fibers, the action potential leaps from node to node (saltatory conduction), which is much faster than continuous conduction in unmyelinated fibers.

Synapses

- A **synapse** is a junction between two neurons or a neuron and an effector.
- **Electrical synapses:** Direct ionic current flow through gap junctions (fast, bidirectional).

- **Chemical synapses:** Involve release of **neurotransmitters** from synaptic vesicles.
 - **Presynaptic neuron:** Sends the signal.
 - **Postsynaptic neuron:** Receives the signal.
 - **Synaptic cleft:** The gap between neurons (20 nm).



Figure 7: Transmission of action potentials across nerve synapses. A: Motor nerve cell body with axon terminals. B: Enlarged synaptic knob releasing neurotransmitter. C: Ultrastructure showing neurotransmitter binding to receptors on postsynaptic membrane to open ion channels.

- Mechanism of chemical transmission (e.g., Acetylcholine):
 1. Action potential arrives at synaptic knob.
 2. Ca^{2+} channels open, Ca^{2+} enters.
 3. Synaptic vesicles fuse with membrane (exocytosis), releasing neurotransmitter.
 4. Neurotransmitter binds to receptors on postsynaptic membrane.
 5. Chemically-gated ion channels open, causing a postsynaptic potential.
 6. Neurotransmitter is inactivated (e.g., by **acetylcholinesterase**) or reabsorbed.
- Synapses can be **excitatory** (depolarizing) or **inhibitory** (hyperpolarizing).
- Other neurotransmitters include **norepinephrine**, **dopamine**, **serotonin**, **GABA** (inhibitory), **glycine** (inhibitory), and **glutamate** (excitatory).



Figure 8: Sequence of events in synaptic transmission at an excitatory synapse. Calcium influx triggers release of acetylcholine, which binds to receptors, opening ion channels and causing postsynaptic membrane excitation. Acetylcholine is then broken down by acetylcholinesterase.

Evolution of Nervous Systems

- **Invertebrates:**
 - **Nerve net:** Simplest system (Cnidarians); diffuse network, no central control.
 - **Bilateral nervous system:** Flatworms have anterior ganglia (brain) and longitudinal nerve cords.
- **Annelids and Arthropods:** Segmental ganglia, ventral nerve cord, more complex brain.
 - In insects, **mushroom bodies** in the brain are associated with learning.
- **Molluscs:** Cephalopods have highly complex brains capable of learning.



Figure 9: Invertebrate nervous systems. A: Nerve net of radiates (Hydra). B: Bilateral system of flatworms (Planaria). C: Segmental ganglia of annelids (Earthworm). D: Segmental system with large ganglia in arthropods (Grasshopper).

- **Vertebrates:**

- **Encephalization:** Great elaboration of the brain.
- **Spinal Cord:** Hollow dorsal nerve cord; enclosed by vertebrae and **meninges**.
 - **Gray matter:** Cell bodies (inner zone).
 - **White matter:** Axon bundles (outer zone).
 - **Dorsal root:** Sensory (afferent).
 - **Ventral root:** Motor (efferent).

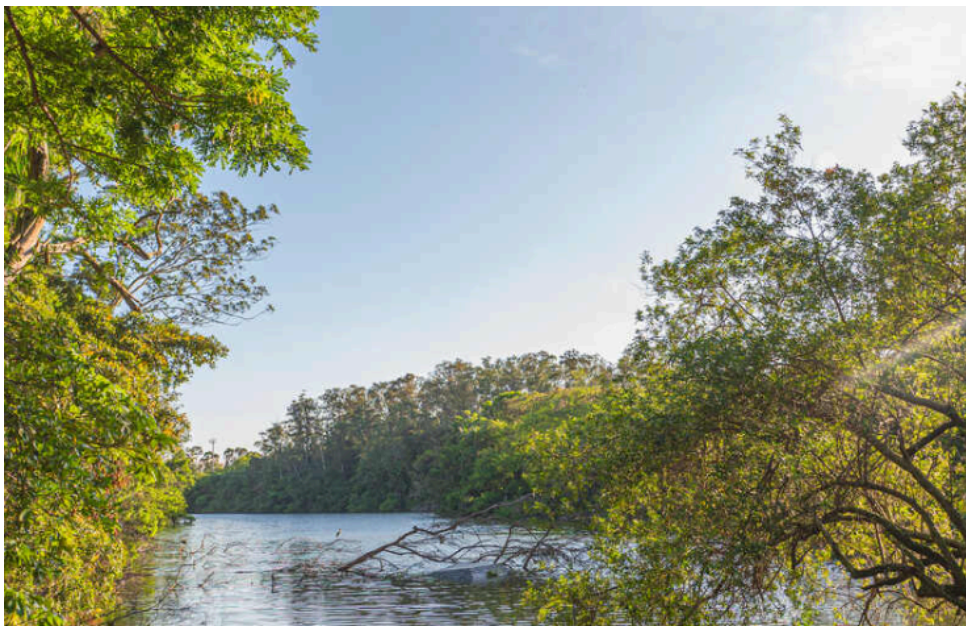


Figure 10: Human spinal cord and its protection. Shows the spinal cord within vertebrae, wrapped in meninges. Dorsal roots carry sensory information, and ventral roots carry motor information.

- **Reflex Arc:**

- Fundamental unit of neural operation.
- Components: Receptor -> Afferent neuron -> CNS (Interneuron) -> Efferent neuron -> Effector.
- **Reflex act:** Involuntary response to a stimulus.



Figure 11: The reflex arc. A: Monosynaptic reflex (knee-jerk) with sensory and motor neurons. B: Multisynaptic reflex involving interneurons, allowing coordination of muscle responses.

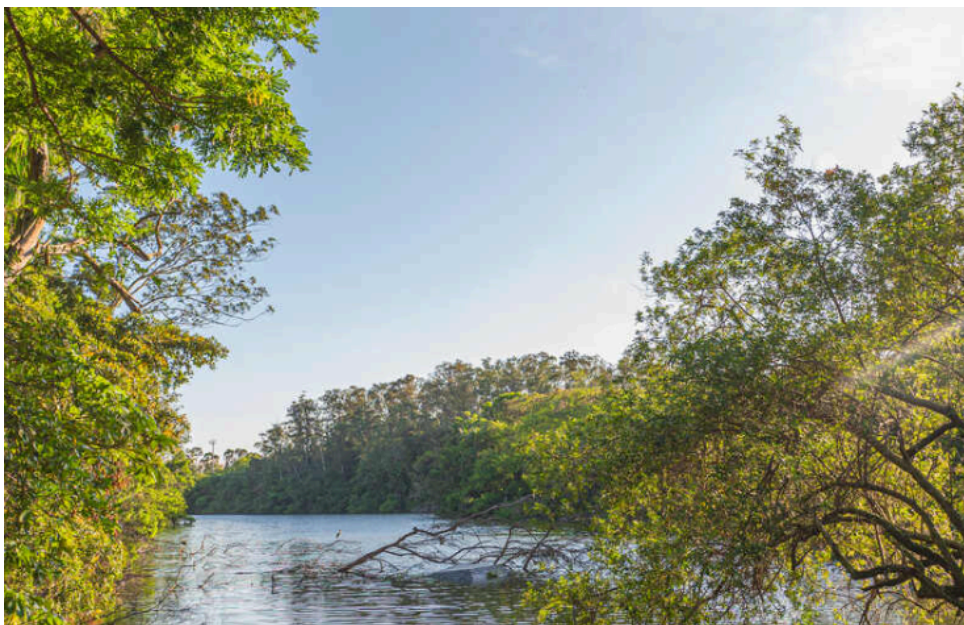


Figure 12: Evolution of the vertebrate brain. Shows the progressive increase in the size of the cerebrum and cerebellum from lampreys to mammals (horse).

- **Brain Divisions:**

- **Hindbrain:**
 - **Medulla oblongata:** Controls vital functions (heartbeat, respiration).

- **Cerebellum**: Controls equilibrium, posture, and movement coordination.
- **Pons**: Connects cerebellum to other brain areas.
- **Midbrain**:
 - **Tectum** (optic lobes): Visual and auditory reflexes.
- **Forebrain**:
 - **Thalamus**: Relay station for sensory information.
 - **Hypothalamus**: Regulates homeostasis (temperature, water balance), endocrine function, and basic drives.



Figure 13: Divisions of the vertebrate brain. Shows the embryonic development from forebrain, midbrain, and hindbrain into adult structures like the cerebrum, thalamus, hypothalamus, optic lobes, cerebellum, and medulla oblongata.

- **Cerebrum**:
 - **Paleocortex** (Limbic system): Emotion, memory (**Hippocampus**), smell.
 - **Neocortex** (Cerebral cortex): Higher mental activities, sensory perception, motor control.



Figure 14: Human brain structure. A: External view showing lobes of the cerebrum (frontal, parietal, temporal, occipital) and cerebellum. B: Midline section showing the corpus callosum, thalamus, hypothalamus, pons, and medulla.

- **Cerebral Cortex:**

- Divided into lobes: Frontal, Parietal, Temporal, Occipital.
- **Motor cortex:** Controls voluntary movement.
- **Sensory cortex:** Conscious perception of touch, pain, etc.
- **Association areas:** Integration, memory, reasoning.
- **Corpus callosum:** Connects left and right hemispheres.
- Left hemisphere: Language, math, logic.
- Right hemisphere: Spatial, artistic, intuitive skills.

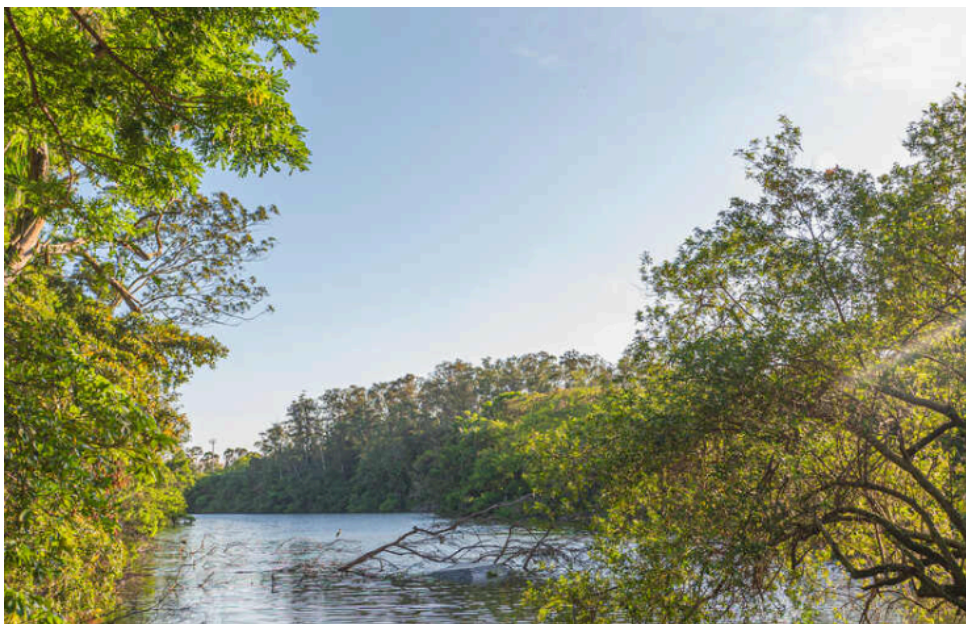


Figure 15: Arrangement of sensory and motor cortices. Maps show the correspondence between cortical areas and the parts of the body they control (motor) or receive sensation from (sensory).

Peripheral Nervous System

- **Somatic Nervous System:** Innervates skeletal muscles (voluntary).
- **Autonomic Nervous System (ANS):** Innervates smooth muscle, cardiac muscle, and glands (involuntary).
 - Two-neuron pathway: **Preganglionic** (CNS to ganglion) and **Postganglionic** (ganglion to effector).



Figure 16: General organization of the autonomic nervous system. Shows the two-neuron pathway: preganglionic neurons from the CNS synapse with postganglionic neurons in ganglia. Sympathetic ganglia are near the cord; parasympathetic are near effectors.

- **Sympathetic Division:** “Fight or flight”; increases heart rate, dilates pupils, inhibits digestion. Neurotransmitter: Norepinephrine (mostly).
- **Parasympathetic Division:** “Rest and digest”; slows heart rate, stimulates digestion. Neurotransmitter: Acetylcholine.



Figure 17: Autonomic nervous system in humans. Sympathetic outflow (thoracolumbar) and parasympathetic outflow (craniosacral). Most organs receive dual innervation with antagonistic effects.

Sense Organs

- **Sensory receptors** are transducers that convert energy (stimulus) into nerve action potentials.
- **Law of specific nerve energies:** The brain interprets signals based on where they arrive (labeled lines).
- Classification:
 - **Exteroceptors:** External environment.
 - **Interoceptors:** Internal organs.
 - **Proprioceptors:** Muscle/joint position.
 - By energy type: Chemoreceptors, Mechanoreceptors, Photoreceptors, Thermoreceptors.

Chemoreception

- Oldest sense; includes **chemotaxis**, **taste** (contact), and **smell** (distance).
- **Pheromones:** Chemical signals for communication within a species.



Figure 18: Pheromone-producing glands of an ant. Ants use numerous chemical signals for communication, including alarm and trail pheromones.

- **Taste:**

- **Taste buds** on tongue papillae.
- Five basic tastes: Sweet, salty, sour, bitter, umami.



Figure 19: Taste receptors. A: Tongue surface with papillae. B: Taste buds on papilla. C: Structure of a taste bud with receptor cells and taste pore.

- **Smell (Olfaction):**

- Olfactory epithelium in nasal cavity.
- Neurons have cilia with receptor proteins.
- Highly sensitive and diverse (thousands of odor receptor genes).
- **Vomer nasal organ (VNO):** Detects pheromones in many vertebrates.



Figure 20: Human olfactory epithelium. Located in the roof of the nasal cavity, it contains olfactory receptor cells with cilia that detect odor molecules.

Mechanoreception

- Sensitive to mechanical forces (touch, pressure, vibration, gravity).
- **Touch:**
 - **Pacinian corpuscles:** Deep pressure receptors; show **adaptation** (respond to change).
- **Pain:**
 - Receptors are free nerve endings.
 - Respond to tissue damage and chemicals like **substance P** and **bradykinins**.
 - **Endorphins** and **enkephalins** are endogenous opiates that block pain.



Figure 21: Response of Pacinian corpuscle to pressure. Mechanical deformation produces a receptor potential. If threshold is reached, an action potential is generated in the nerve fiber.

- **Lateral-Line System:**

- In fishes and aquatic amphibians.
- Detects water vibrations and currents.
- **Neuromasts**: Receptor units with hair cells in a **cupula**.



Figure 22: Lateral-line system. A: Location on a bony fish. B: Structure of a neuromast with hair cells embedded in a cupula, detecting water vibrations.

- **Hearing:**

- Invertebrates: Tympanic organs (e.g., insects). Moths have ears to detect bat echolocation.

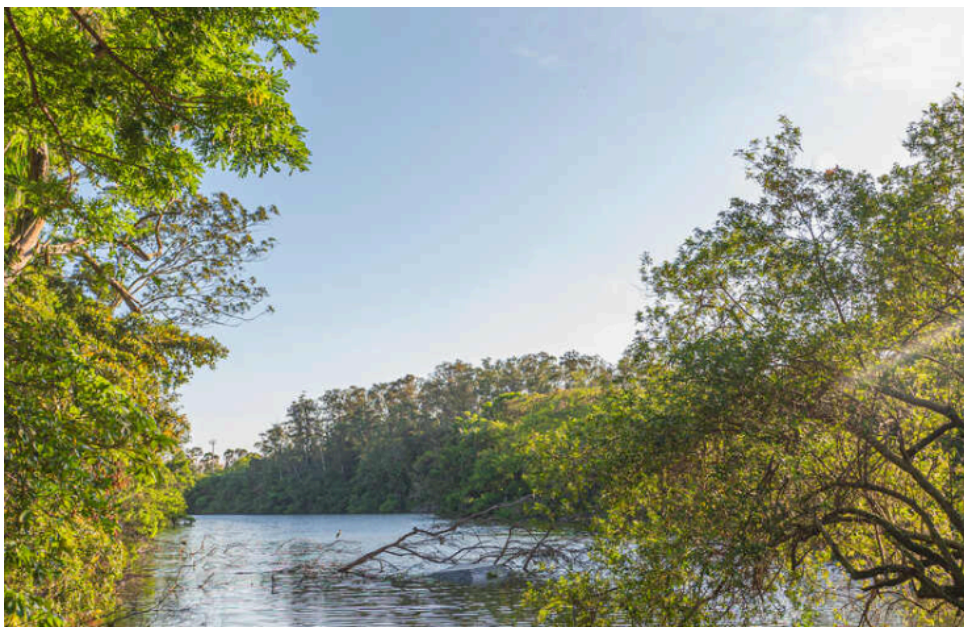


Figure 23: Ear of a moth. Contains A1 and A2 receptors to detect ultrasonic cries of bats, enabling evasive maneuvers.

- Vertebrates: Ear originated as a balance organ (**labyrinth**).
- Fish have **otoliths** (saccule/utricle) and semicircular canals.



Figure 24: Vestibular apparatus of a teleost fish. Includes semicircular canals for angular acceleration, and utricle, saccule, and lagena for balance and sound reception.

- Mammalian Ear:
 - **Outer ear:** Pinna and auditory canal.
 - **Middle ear:** Tympanic membrane and ossicles (**malleus, incus, stapes**). Amplifies sound. Connected to pharynx by **Eustachian tube**.
 - **Inner ear:** **Cochlea** (hearing) and vestibular apparatus (balance).
 - **Organ of Corti:** Contains hair cells on the **basilar membrane**; the actual sensory apparatus.

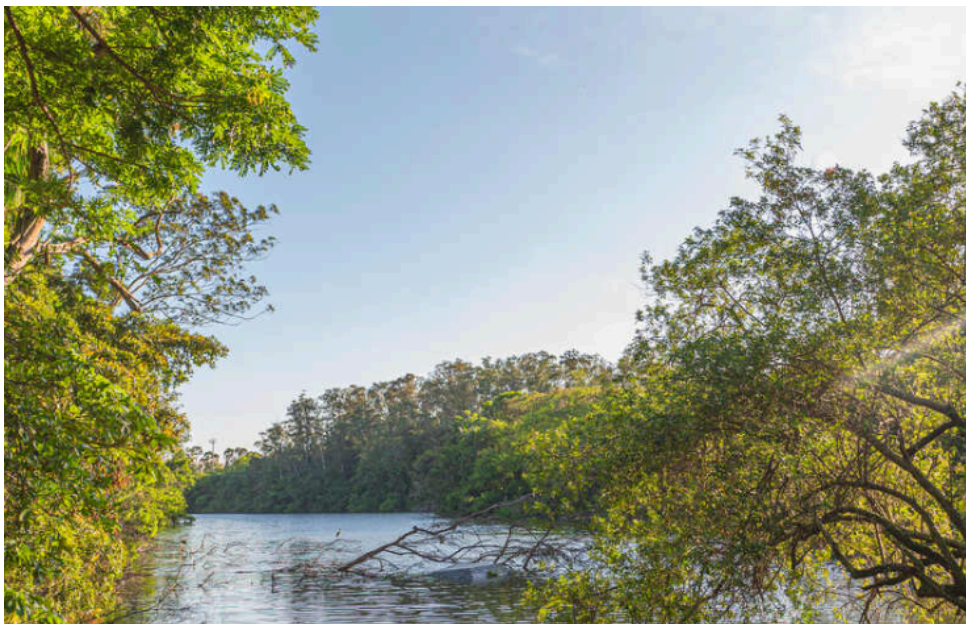


Figure 25: Human ear. A: External, middle, and inner ear. B: Middle ear ossicles and inner ear cochlea. C: Cross section of cochlea showing organ of Corti. D: Ultrastructure of organ of Corti with hair cells.

- **Pitch discrimination: Place hypothesis.** Different frequencies resonate different parts of the basilar membrane.



Figure 26: Frequency localization in the cochlea. High frequencies resonate the basilar membrane near the oval window; low frequencies travel further to the apex.

- **Traveling Waves:** Sound waves cause the basilar membrane to vibrate, with the wave reaching maximum amplitude at a specific point determined by frequency.

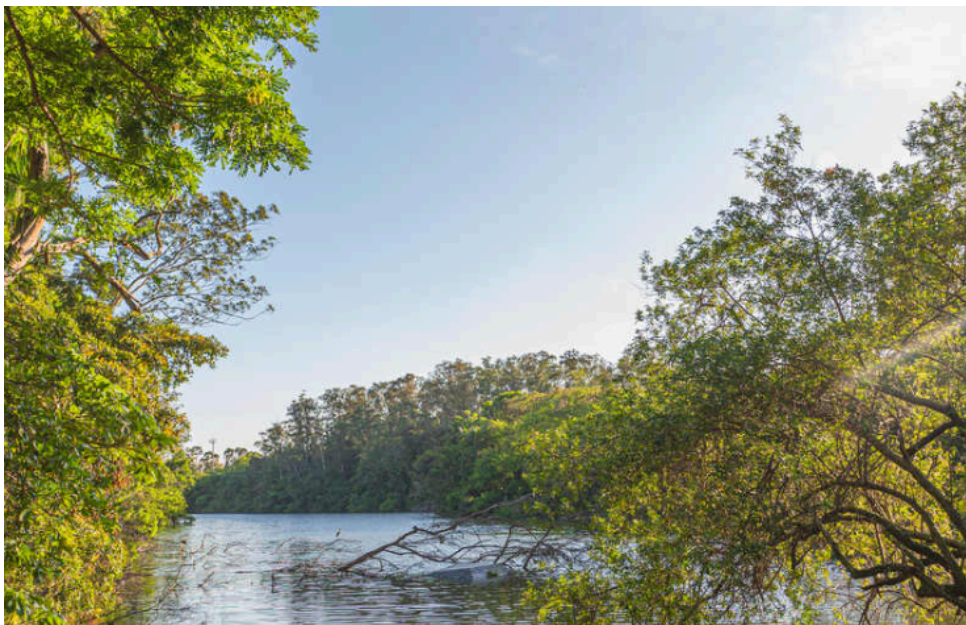


Figure 27: Traveling waves along the basilar membrane. Sound waves cause the membrane to vibrate, stimulating hair cells at the point of maximum amplitude corresponding to the frequency.

- **Equilibrium:**
 - **Statocysts:** Invertebrate gravity sensors (statolith + hair cells).



Figure 28: Statolith of the crayfish. A static balance organ where a heavy statolith stimulates hair cells to indicate position relative to gravity.

- Vertebrate Vestibular Organ:
 - **Utricle and Saccule**: Static balance (gravity).
 - **Semicircular Canals**: Rotational acceleration. **Ampulla** contains **cupula** and hair cells.



Figure 29: Response of semicircular canals to angular acceleration. Inertia of endolymph bends the cupula, stimulating hair cells to detect rotation.

Photoreception (Vision)

- **Photoreceptors** range from simple eyespots to complex eyes.



Figure 30: Eyespot of the dinoflagellate *Nematodinium*. A complex single-celled structure with a lens and pigment cup.

- **Compound Eye** (Arthropods):
 - Composed of **ommatidia**.
 - Excellent for motion detection; mosaic image.



Figure 31: Compound eye of an insect. Composed of many ommatidia, each with a lens and reticular cells, forming a mosaic image.

- **Camera-type Eye** (Vertebrates, Cephalopods):
 - **Cornea**: Refracts light.
 - **Lens**: Focuses light (accommodation).
 - **Iris/Pupil**: Regulates light entry.
 - **Retina**: Light-sensitive layer.



Figure 32: Structure of the human eye. A camera-type eye with cornea, lens, iris, and retina. The fovea centralis is the area of keenest vision.

- **Retina Structure:**

- **Rods:** Dim light, black and white. High sensitivity, low acuity.
- **Cones:** Bright light, color. Concentrated in **fovea centralis**. Low sensitivity, high acuity.
- Intermediate neurons: Bipolar, horizontal, and amacrine cells process signals before ganglion cells.

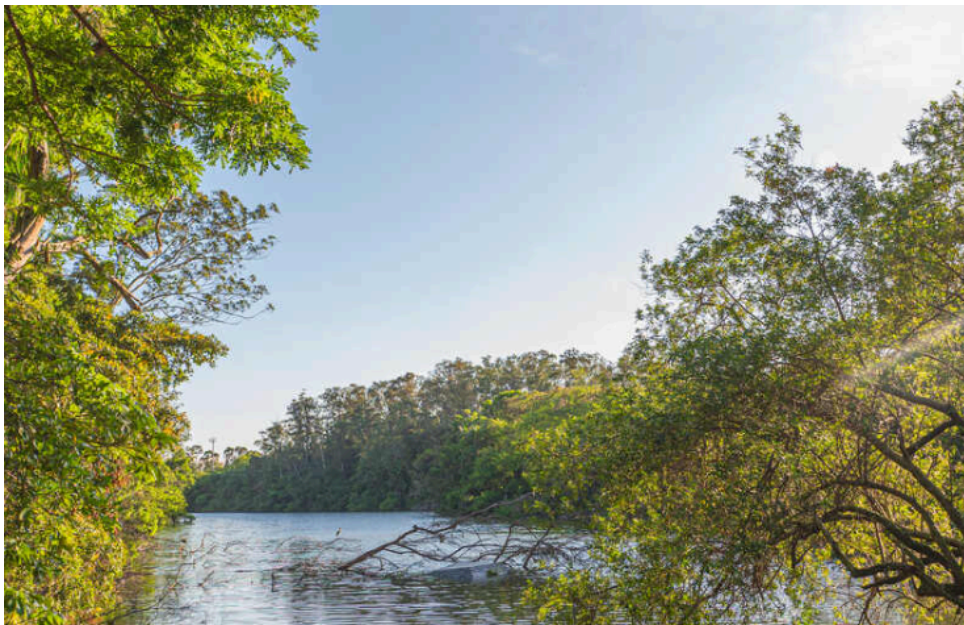


Figure 33: Structure of a primate retina. Light passes through layers of neurons (ganglion, bipolar) to reach photoreceptors (rods and cones) at the back.

- **Chemistry of Vision:**

- **Rhodopsin** (opsin + retinal) is the visual pigment.
- Light causes retinal isomerization -> opsin activation -> signal cascade -> hyperpolarization.

- **Dark adaptation:** Regeneration of rhodopsin in the dark takes time (30 mins).
- **Color Vision:**
 - Trichromatic theory: Three types of cones (Blue, Green, Red).



Figure 34: Absorption spectrum of human vision. Three types of cones (blue, green, red) respond maximally to different wavelengths, enabling color vision.